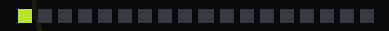
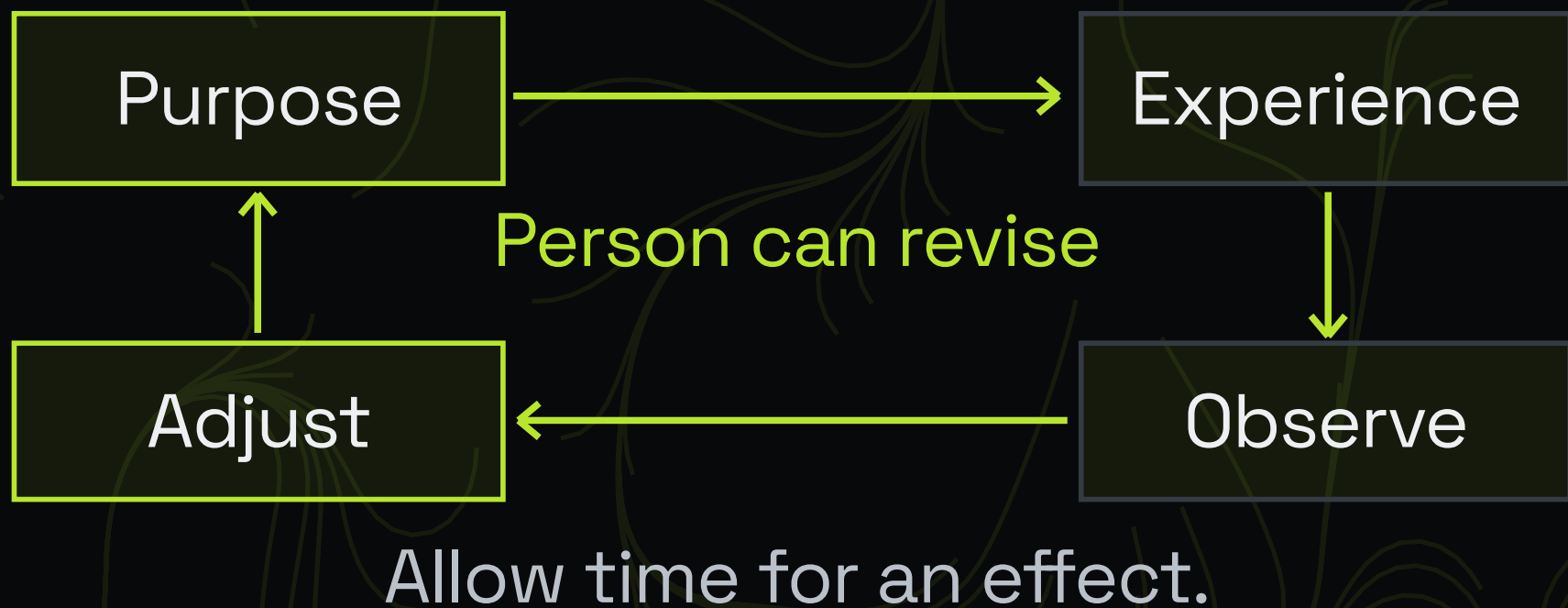


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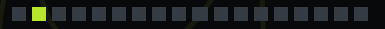


Draw the feedback.

Before calling software adaptive, draw the feedback



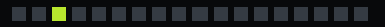
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The design question

“Adapt to the learner” sounds like a product requirement. It leaves the most important choice unstated: what should remain worthwhile as the learner and the situation change? A learning tool might aim to help someone make progress while encountering useful difficulty. That is an intention. It still needs a way to observe the experience, interpret what is happening and offer a change the person actually wants.

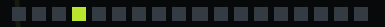
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The design question

I would begin by drawing those relationships. The diagram is more informative than assigning the product an adaptive label because it reveals what the system can observe and what it can change.

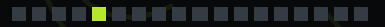
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Name the condition, then the observation

Consider fast task completion. It might indicate understanding. It might indicate a trivial task, or a person rushing through an explanation they do not follow. Completion time alone cannot choose between those interpretations.

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Name the condition, then the observation

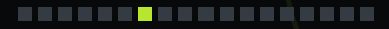
Give the example another observation: the learner reports confusion. A plausible next step is to offer a different explanation and let them decide whether to continue or change the challenge. Immediately making the task harder would depend on an interpretation the evidence does not support.

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Name the condition, then the observation

This is a proposed interface response, not a validated educational controller. The value of the example is the question it exposes: which observations can disagree, and what may the product do when they do?

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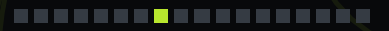
Name the condition, then the observation



Allow time for an effect.

Figure 1. A conceptual loop connecting purpose, experience, observation and adjustment. The person can revise the purpose; the diagram does not prove a stable or effective controller.

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Biology makes the mechanism interesting

Biological homeostasis provides a reason to study regulation seriously. A particularly precise example is antithetic integral feedback. Briat, Gupta and Khammash analyze a biochemical controller whose paired species encode an integral action through their difference. Their results concern specified stochastic reaction networks and conditions, including the behavior of the closed loop.

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Biology makes the mechanism interesting

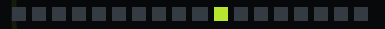
They are not a general theorem about AI agents. Briat, Gupta and Khammash, 2016.

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Biology makes the mechanism interesting

The general engineering idea of integral action is accessible without reproducing the biochemical mechanism: retain accumulated error, so a persistent deviation continues to influence the response. In a simple discrete illustration, $I_{\text{next}} = I_{\text{current}} + e \cdot \Delta t$. Here e is the target minus the observed value, Δt is the time step and I accumulates error over time. If e is measured in units U and time in seconds, I has units $U \cdot \text{seconds}$.

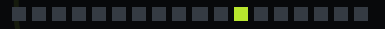
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Biology makes the mechanism interesting

This is a conceptual discretization, not the paper's molecular implementation. The equation also makes a limitation visible. Accumulating an error does not, by itself, establish that the resulting system is stable, that the target is reachable or that the observation is meaningful. The response has to be designed for the dynamics and the available actions.

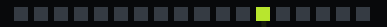
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Keep time in the picture

An adjustment can take time to have an effect. If a product changes its behavior again before that effect can be observed, the next reading can be difficult to interpret. A person may also need time to settle into a new example or learning activity.

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Keep time in the picture

Put that delay into the design discussion. What changes immediately? What can be judged only after the next attempt? When should the product wait, ask or offer another adjustment? These questions can produce different interaction choices even when the model generates excellent material.

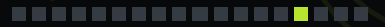
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Keep time in the picture



Figure 2. Fast completion and reported confusion are considered together. Offering another explanation is a design proposal; the diagram contains no measured learning result.

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The person chooses what is worth sustaining

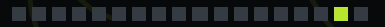
A learner preparing for a task tomorrow may want a different experience from someone exploring a subject for pleasure. Keeping a difficulty score steady would not resolve that difference. The purpose belongs in the design, and the person needs a way to change it.

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The person chooses what is worth sustaining

That is where I would resist an automatic controller analogy. People can reinterpret the task, reject the objective or want to explore something the current measurements do not represent. Make those changes expressible rather than treating them as disturbances to suppress.

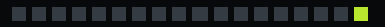
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The person chooses what is worth sustaining

For a prototype, write five items on one page: the condition you hope to sustain, observations that can disagree, available adjustments, the time needed to see an effect and the choices that remain with the person. Work through a changed situation before adding more machinery.

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The person chooses what is worth sustaining

The result may justify an adaptive policy. It may instead reveal a useful control the person should operate directly. Either outcome gives the team something concrete to build. The ambition is to make the experience remain useful while it changes; the diagram shows what that ambition asks of the system.